

DEFINITION — HOW DATA IS DESCRIBED

API INTERFACES5

**ODCS**
BITOL @ LF

**OpenAPI**
LF

**AsyncAPI**
LF

**GraphQL**
LF

**gRPC**
CNCF

DATA PRODUCTS4

**ODPS**
BITOL @ LF

**ODPSpec**
LF

**DPDS**
ODM

**DPROD**
OMG

SCHEMA5

**XML Schema**
W3C

**JSON Schema**
IETF

**SQL DDL**
ISO/IEC

**AVRO Schema**
ASF

**Protobuf**
GOOGLE

SEMANTICS6

**OSI**
OSI INITIATIVE

**RDF/OWL**
W3C

**DCAT**
W3C

**SKOS**
W3C

**SHACL**
W3C

**JSON-LD**
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS6

**CSV**
IETF

**JSON**
IETF

**XML**
W3C

**PARQUET**
ASF

**AVRO**
ASF

**ORC**
ASF

OPEN TABLE FORMATS3

**Iceberg**
ASF

**Delta**
LF

**Hudi**
ASF

STORAGE SYSTEMS2

**S3**
AWS

**HDFS**
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3

**JDBC**
JCP

**ODBC**
ISO/IEC

**ADBC**
ASF

INTERCONNECTION2

**HTTP**
IETF

**FTP / SFTP**
IETF

MESSAGING4

**Kafka**
ASF

**AMQP**
OASIS

**MQTT**
OASIS

**CloudEvents**
CNCF

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

**Apache Arrow**
ASF

**DataFrame API**
DATA-APIS

PROCESSING4

**dbt**
DBT LABS

**Spark**
ASF

**Pandas**
NUMFOCUS

**Ibis**
IBIS

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5



Iceberg Catalog

ASF



Unity Catalog

LF



Hive Metastore

ASF



DuckLake

DUCKDB LABS



Schema Registry

CONFLUENT

LINEAGE1



OpenLineage

LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2



SQL

ISO/IEC



SPARQL

W3C

DATA QUALITY3



dbt

DBT LABS



Great Expectations

GX LABS



SodaCL

SODA

OBSERVABILITY2



Open Telemetry

CNCF



OORS

BITOL @ LF

POLICIES1



OPA

CNCF

AI INTERFACES2



MCP

LF



A2A

LF

☐ Open standard

☒ Highlighted by Entropy Data — the ones to bet on today

☐ Emerging / draft

☐ Single-vendor — muted